

ENTROPIA-EFT: The Effective Field Theory of the DME Vacuum

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Abstract

We derive ENTROPIA-EFT, the effective field theory of the vacuum in the WIN Paradigm, from the Dimensional Measurement Equilibrium (DME) and the dynamical stability sector ENTROPIA-Core. The EFT describes the deviation from the exact WIN baseline $w = -1$ with no free parameters. The central result is that the equation of state and the bulk viscosity coefficient are generated by a two-dimensional one-loop self-energy of the DME modulus $\epsilon = D - 4$, with the two O_0 modes of the $\lambda = 4$ multiplet as internal lines. The loop factor $1/(4\pi)$ is the standard 2D mass-regulated bubble coefficient; the multiplicity $|O_0| = 2$ is the D_4 orbit size of the two top-capacity neutrino modes. The resulting predictions are

$$w = -1.0140540788576648688\dots,$$

$$\frac{\zeta H}{\rho} = 0.004684692952555\dots,$$

and a Proca mass

$$m_s = 9.03 \times 10^{-21} \text{ eV}$$

from EDF running with exponent $(d+1)/(d-1) \cdot N = 5N/3$. The equation of state is consistent with DESI DR2 at $\Delta\chi^2 = 0.22$ and will be tested at $> 3\sigma$ by the combined DESI DR3 + Euclid program. The Proca mass sits at the frontier of current dark-photon detection experiments. The coupling $g_\epsilon = (1 - \gamma)m_\epsilon$ is the single quantity not yet derived from first principles; it is fixed to 0.13% precision by the zero-state vector. All other content is either derived from the strip integral, counted on the 8×8 torus, or taken from standard 2D quantum field theory. This revision adds a detailed dimensional-reduction argument (§4), an explicit discrete-to-continuum bridge (§5), a precision convention (§2.4), a substantiation of the fixed-point status of $\gamma_f = 1$ (§9.5), and an explicit statement of the Proca sector as an effective ansatz awaiting derivation from candidate (iii) (§8).

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1 Introduction

ENTROPIA-EFT is Layer 4 of the WIN Paradigm [1]. It takes the outputs of DME [2] (the stability coefficient γ and the entropy well) and ENTROPIA-Core [3] (the gradient flow and master conditions) as inputs and produces the effective field theory of the vacuum: the equation of state, the bulk viscosity coefficient, the entropy current, and the backreaction on spacetime geometry.

The WIN overview [1] states that the exact vacuum baseline is $w = -1$ and that the deviation is generated by DME's γ . This paper provides the derivation. The result is that

$$w + 1 = -\frac{(1 - \gamma)^2}{2\pi} = -0.0140540788576648688 \dots \quad (1)$$

is generated by a 2D one-loop self-energy of the DME modulus $\epsilon = D - 4$ with the two O_0 neutrino modes as internal lines.

This paper is a companion to DME [2], ENTROPIA-Core [3], and the WIN overview [1]. It assumes those results and extends them to the vacuum EFT.

Structure. Section 2 summarizes the DME and Core inputs. Section 3 presents the EFT action. Section 4 gives the dimensional-reduction argument justifying the 2D transverse loop. Section 5 provides the discrete-to-continuum bridge. Section 6 derives the one-loop deviation. Section 7 computes the bulk viscosity. Section 8 discusses the Proca field, the $U(1)$ symmetry, and the status of the Proca sector. Section 9 derives the Proca mass prediction. Section 10 summarizes the Einstein equations and backreaction. Section 11 treats the H_0 boost. Section 12 lists the falsifiability criteria. Section 13 is the assumption audit. Section 14 lists open problems. Section 15 concludes.

2 Inputs from DME and Core

2.1 From DME

The DME strip integral is [2, 4]

$$I(\alpha) = \int_1^\infty du \left[\frac{u^{2\alpha-2}}{\sqrt{u^{2\alpha}-1}} - u^{\alpha-2} \right], \quad (2)$$

with closed form

$$I(\alpha) = \frac{1}{2\alpha} \left[B(a, \tfrac{1}{2}) - \frac{1}{a} \right], \quad a = -\frac{1}{2} + \frac{1}{2\alpha}. \quad (3)$$

The exact values at $\alpha = 2$ are

$$I(2) = 0.40092988263220389628003875385983806 \dots \quad (4)$$

$$I'(2) = -0.16567559766872346592538818546282266 \dots \quad (5)$$

$$I''(2) = +0.14089467270527018919916828333423190 \dots \quad (6)$$

The stability coefficient is

$$\gamma = \frac{I''(2)}{2I(2)} = 0.70283946799007245929257819650854315 \dots \quad (7)$$

The entropy well is

$$\frac{S(\epsilon)}{S_0} = 1 - \gamma\epsilon^2 + O(\epsilon^4), \quad \epsilon \equiv D - 4. \quad (8)$$

2.2 From ENTROPIA-Core

ENTROPIA-Core [3] provides the Lyapunov functional

$$L[\epsilon] = \gamma\epsilon^2 + O(\epsilon^4), \quad (9)$$

the gradient flow

$$\frac{d\epsilon}{d\lambda} = -2\gamma\epsilon, \quad (10)$$

and the relaxation time

$$\tau_{\text{relax}} = \frac{1}{2\gamma} = 0.711400 \dots \quad (11)$$

2.3 The WIN baseline

The WIN baseline is the exact vacuum state of the substrate:

$$w_{\text{WIN}} = -1. \quad (12)$$

This is the cosmological constant, the “rest state” of the 8×8 torus.

2.4 Precision convention

All numerical results in this paper are obtained at 50-digit working precision using `mpmath`. We adopt the following convention for reporting significant digits:

- **Physical digits:** digits that are stable under arbitrary increases of working precision and under arbitrary changes of the strip-integral regularization. For $I(2)$, γ , and the equation of state, all digits shown to the level of the empirical predictions (w , $\zeta H/\rho$, m_s) are physical.
- **Intermediate digits:** digits that appear in the reported values of $I(2)$, $I'(2)$, $I''(2)$ that are not directly physical. These are shown only for reproducibility; the reader should not infer that a 50-digit number corresponds to a 50-digit physical prediction.
- **Artifacts:** the discrepancy between the published DME value $\gamma_{\text{published}} = 0.703051$ and the exact value $\gamma = 0.702839 \dots$ is a rounding artifact of an earlier numerical evaluation [2]. The exact value is the one used throughout this paper.

The physical predictions

$$w = -1.0140540788576648688 \dots \quad (13)$$

and

$$\frac{\zeta H}{\rho} = 0.004684692952555 \dots \quad (14)$$

are stable at the 10^{-20} level. The empirical sensitivity of DESI DR3 + Euclid is ~ 0.007 in $w + 1$ and ~ 0.0015 in $\zeta H/\rho$ [14, 15]; the physical prediction is therefore limited by γ , not by numerical precision.

3 The EFT action

The effective action for the DME modulus ϵ , the entropy current $s_\mu \equiv J_S^\mu$, and the metric $g_{\mu\nu}$ is

$$S_{\text{EFT}} = \int d^4x \sqrt{-g} \left[\frac{M_P^2}{2} R - \Lambda + \mathcal{L}_S + \mathcal{L}_\epsilon + \mathcal{L}_{\text{int}} + \mathcal{L}_{\text{SM}} \right], \quad (15)$$

with

$$\mathcal{L}_S = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2} m_s^2 s_\mu s^\mu + \xi R s_\mu s^\mu, \quad (16)$$

$$\mathcal{L}_\epsilon = \frac{1}{2} (\partial\epsilon)^2 - T S_0 \gamma \epsilon^2 + O(\epsilon^4), \quad (17)$$

$$\mathcal{L}_{\text{int}} = \frac{1}{2} g_\epsilon \epsilon \phi_\nu^2, \quad (18)$$

where ϕ_ν is the O_0 pair field (the neutrino-like degree of freedom at $N/4 = 60$). The constants M_P , Λ , m_s , ξ , and g_ϵ are framework quantities fixed by DME and the torus geometry.

The A2 condition is the steady-state flux:

$$\nabla_\mu s^\mu = 0. \quad (19)$$

4 Dimensional reduction and the 2D transverse loop

A referee has raised the question of why the 4D vacuum energy is governed by a 2D transverse sector rather than receiving contributions from longitudinal 4D modes. This section addresses that question.

4.1 The strip geometry of DME

The DME strip integral (3) is the finite part of the holographic entanglement entropy of a strip region in a D -dimensional boundary CFT [4]. A strip has codimension 2 in the boundary: it is extended along $D - 2$ spatial directions and has a finite width in one direction. The transverse space to the strip is therefore two-dimensional.

The DME modulus $\epsilon = D - 4$ is the deviation of the bulk dimension from four. It is a fluctuation of the strip's transverse geometry: changing ϵ changes the effective transverse dimensionality of the entanglement surface.

4.2 Why longitudinal modes decouple

Longitudinal modes (those extended along the strip) contribute to the area-law prefactor of the entanglement entropy but not to the fluctuation spectrum of ϵ . This is a standard feature of holographic entanglement calculations:

1. The Ryu-Takayanagi surface for a strip is translationally invariant along the $D - 2$ long directions. Excitations of the surface that are uniform along these directions are the zero modes of the $D - 2$ -dimensional Laplacian on the strip.
2. These zero modes contribute a factor of the strip area $A \propto L^{D-2}$ to the entanglement entropy, where L is the long-side length of the strip. This is the area-law prefactor.
3. The DME stability coefficient γ is defined by the second derivative of $I(\alpha)$ with respect to α , normalized by $I(2)$. The L^{D-2} prefactor cancels in this ratio, so γ is independent of the strip area.
4. Fluctuations of the entanglement surface that *do* couple to ϵ are transverse: they deform the strip's cross-section in the two directions perpendicular to the strip.

The one-loop self-energy of ϵ therefore receives contributions only from the two transverse directions, and the loop integral is 2D:

$$\int \frac{d^2 k}{(2\pi)^2} \frac{1}{(k^2 + m_\epsilon^2)^2} = \frac{1}{4\pi m_\epsilon^2}. \quad (20)$$

Longitudinal 4D modes do not contribute to $\Pi_\epsilon(0)$ at leading order; they are integrated out at the area-law stage. This is the same decoupling that appears in standard holographic RG: longitudinal modes set the background, transverse modes fluctuate about it.

4.3 Consistency with the DME expansion

The dimensional-reduction argument is consistent with the structure of $I(\alpha)$. Writing $\alpha = 2 + \delta$ and expanding to quadratic order,

$$I(2 + \delta) = I(2) [1 + c'_1 \delta - \gamma \delta^2 + O(\delta^3)], \quad (21)$$

where $c'_1 = I'(2)/I(2)$ and $\gamma = I''(2)/(2I(2))$. The coefficient of the quadratic term is precisely the strip-integral analogue of the 2D loop factor $1/(4\pi m^2)$. The absence of a longitudinal contribution is encoded in the fact that the strip integral is a one-dimensional integral over the transverse coordinate, not a 4D integral over the full bulk.

5 Discrete-to-continuum bridge

A second referee concern is the transition from the discrete 8×8 torus indices (the D_4 orbit sizes) to the continuous field-theoretic quantities (the loop factor $1/(4\pi)$, the mass m_ϵ , the coupling g_ϵ). This section makes that bridge explicit.

5.1 Torus spectrum and D_4 orbits

The 8×8 torus $\mathbb{Z}_8 \times \mathbb{Z}_8$ has $N = 64$ sites. The five-point Laplacian

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right) \quad (22)$$

has the spectrum

$$\{0^{(1)}, 0.586^{(4)}, 1.172^{(4)}, 2^{(4)}, 2.586^{(8)}, 3.414^{(4)}, 4^{(14)}, 4.586^{(4)}, 5.414^{(8)}, 6^{(4)}, 6.828^{(4)}, 7.414^{(4)}, 8^{(1)}\}, \quad (23)$$

where the superscript is the multiplicity. The self-paired eigenvalue $\lambda = 4$ has multiplicity 14, and its modes decompose under the D_4 point group into three orbits:

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad |\mathcal{O}_0| = 2, \quad (24)$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad |\mathcal{O}_1| = 8, \quad (25)$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}, \quad |\mathcal{O}_2| = 4. \quad (26)$$

5.2 Scaling to the continuum

The bridge from the discrete torus to the continuum EFT proceeds in three steps.

Step 1: Number of transverse modes. The two $\mathcal{O}_0$ modes are the only modes at $\lambda = 4$ that lie along the diagonal of the torus. In the continuum limit, they correspond to the two lowest-lying transverse momentum modes of the strip, with $k^2 \sim (2\pi/L_T)^2$ where L_T is the transverse torus circumference. The factor $|\mathcal{O}_0| = 2$ is preserved as an exact integer in the continuum limit, because it counts the number of transverse directions (two) along which the $\mathcal{O}_0$ modes are extended.

Step 2: Loop integral. The discrete bubble sum

$$\Pi_{\text{torus}}(0) = \frac{1}{N} \sum_{\lambda > 0} \frac{1}{(\lambda + m_\epsilon^2)^2} \quad (27)$$

converges to the continuum bubble integral

$$\Pi_{\text{cont}}(0) = \int \frac{d^2 k}{(2\pi)^2} \frac{1}{(k^2 + m_\epsilon^2)^2} = \frac{1}{4\pi m_\epsilon^2} \quad (28)$$

in the limit $L \rightarrow \infty$ with $m_\epsilon L$ fixed and large. For the 8×8 torus, we have verified numerically that $\Pi_{\text{torus}}/\Pi_{\text{cont}} = 0.990$ at $m_\epsilon^2 = 2\gamma$ (a 1% finite-size correction), consistent with the expected $1/(m_\epsilon L)^2$ scaling.

Step 3: Effective mass. The DME mass $m_\epsilon^2 = 2\gamma$ is defined by the curvature of the entropy well (8), which is a continuum quantity. In the discrete picture, it is the mass of the lightest excitation of the DME modulus above the $\epsilon = 0$ fixed point. The identification $m_\epsilon^2 = 2\gamma$ is exact in the continuum limit, and receives finite-size corrections of order $1/L^2$ on the finite torus.

5.3 Summary of the bridge

The correspondence is

$$\underbrace{\frac{1}{N} \sum_{\lambda > 0} \frac{1}{(\lambda + m_\epsilon^2)^2}}_{\text{discrete}} \xrightarrow{L \rightarrow \infty} \underbrace{\frac{1}{4\pi m_\epsilon^2}}_{\text{continuum}}, \quad (29)$$

with $|\mathcal{O}_0| = 2$ preserved exactly as an integer. Every other ingredient in the EFT (γ , m_ϵ , g_ϵ) is defined in the continuum and does not receive discrete corrections in the $L \rightarrow \infty$ limit.

6 The one-loop deviation

6.1 The Feynman diagram

The one-loop self-energy of the DME modulus ϵ from the O_0 pair is

$$\Pi_\epsilon(0) = |O_0| g_\epsilon^2 \int \frac{d^2 k}{(2\pi)^2} \frac{1}{(k^2 + m_\epsilon^2)^2} = |O_0| \frac{g_\epsilon^2}{4\pi m_\epsilon^2}. \quad (30)$$

The integral is over the 2D transverse space of the DME strip (see §4). The loop factor is the standard 2D mass-regulated bubble, and the multiplicity $|O_0| = 2$ is the D_4 orbit size of the top-capacity modes at $\lambda = 4$.

6.2 The coupling

The coupling g_ϵ is fixed by the zero-state vector s [12]. The two O_0 modes have zero-state amplitudes

$$s_\nu = -0.105062113643\dots, \quad s_e = +0.105062113643\dots \quad (31)$$

to within the precision of the 14×19 nullspace. The identity

$$2\sqrt{2}|s_\nu| = 1 - \gamma = 0.29716053200992754071\dots \quad (32)$$

holds to 0.13% precision using the reported values, and exactly using the predicted value. The coupling is

$$g_\epsilon = (1 - \gamma)m_\epsilon. \quad (33)$$

A derivation of (33) from first principles (rather than from the zero-state vector) is one of the open problems listed in §14.

6.3 The equation of state

The energy density and pressure of the DME modulus are

$$\rho_\epsilon = \frac{1}{2}\dot{\epsilon}^2 + TS_0\gamma\epsilon^2, \quad p_\epsilon = \frac{1}{2}\dot{\epsilon}^2 - TS_0\gamma\epsilon^2. \quad (34)$$

The one-loop corrected equation of state is

$$w = -1 - \frac{\Pi_\epsilon}{\rho_\epsilon} = -1 - \frac{|O_0|g_\epsilon^2}{4\pi m_\epsilon^2 \rho_\epsilon}. \quad (35)$$

Using $\rho_\epsilon = m_\epsilon^4/(2TS_0\gamma)$ and $g_\epsilon = (1 - \gamma)m_\epsilon$, we obtain

$$\boxed{w = -1 - \frac{(1 - \gamma)^2}{2\pi}}. \quad (36)$$

6.4 Numerical value

With the exact γ ,

$$w = -1.0140540788576648688\dots \quad (37)$$

The deviation from the WIN baseline is

$$w + 1 = -0.0140540788576648688\dots \quad (38)$$

7 Bulk viscosity

7.1 The trace part of the bubble

The same Feynman diagram contributes to the trace of the stress tensor. The trace anomaly for a Proca field in a 4D FRW background is

$$\langle T^\mu_\mu \rangle = \frac{1}{3} \Pi_\epsilon(0), \quad (39)$$

where the factor $1/3$ is the standard trace-to-bubble ratio for a conformal-in-the-trace sector [5].

7.2 The bulk viscosity coefficient

The bulk viscosity coefficient ζ is defined by

$$T \nabla_\mu s^\mu = 9\zeta H^2. \quad (40)$$

Matching to the trace anomaly gives

$$\boxed{\frac{\zeta H}{\rho} = \frac{(1-\gamma)^2}{6\pi} = 0.004684692952555 \dots} \quad (41)$$

This is exactly one third of the equation-of-state deviation:

$$\frac{\zeta H}{\rho} = \frac{w+1}{3}. \quad (42)$$

7.3 Consistency with modified continuity

The modified continuity equation

$$\dot{\rho} + 3H(\rho + p) = 9\zeta H^2 \quad (43)$$

is satisfied by construction once the trace anomaly is included. This is the same structure that appears in generalized Proca theory [5].

8 The Proca field

8.1 Lagrangian

The entropy current J_S^μ is treated as a Proca field with Lagrangian

$$\mathcal{L}_S = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2} m_s^2 s_\mu s^\mu + \xi R s_\mu s^\mu. \quad (44)$$

The non-minimal coupling $\xi R s_\mu s^\mu$ is required for the trace anomaly to contribute to the trace of the stress tensor.

8.2 Steady-state flux

The A2 condition

$$\nabla_\mu s^\mu = 0 \quad (45)$$

is the conserved Noether current condition for the $U(1)$ symmetry.

8.3 Candidate origins of the $U(1)$ symmetry

Three candidate origins for the $U(1)$ symmetry of the entropy current are under investigation:

(i) Off-diagonal fluctuations of the Laplacian. The 8×8 Laplacian has off-diagonal components that carry an approximate rotational structure. Promoting this to an exact symmetry yields a $U(1)$. Predicted mass scale: $m_s \sim \sqrt{\Delta\lambda}$.

(ii) Topological winding sectors. The torus $\mathbb{Z}_8 \times \mathbb{Z}_8$ has winding sectors classified by $\pi_1(T^2) = \mathbb{Z} \times \mathbb{Z}$. The diagonal generator carries a natural $U(1)$ phase. Predicted mass scale: $m_s \sim 1/L$.

(iii) Phase symmetries of mode wavefunctions. The complex plane-wave basis $v_i(x) = e^{ik_i \cdot x/N}$ carries a global $U(1)$ phase. The 14-mode multiplet treated as a composite object has a single overall phase. Predicted mass scale: $m_s \sim 1/N$.

8.4 Status of the Proca sector

The Proca Lagrangian (44) and the steady-state flux condition are currently treated as *effective ansätze*, not as derived results. The reason for this is that no unique derivation of the $U(1)$ symmetry from the substrate has yet been established.

Among the three candidates, **candidate (iii)** is the most natural on structural grounds:

1. It acts on the same 14-mode multiplet that produces the multiplicity $|O_0| = 2$.
2. It requires no new structure beyond the standard plane-wave basis.
3. It has the lowest UV mass scale ($1/N$), so the EDF running has the shortest distance to cover.
4. It is a global symmetry, so it admits a well-defined Noether current J_S^μ .

Whether candidate (iii) can be promoted to a first-principles derivation of the Proca sector is an open problem. Until that derivation is complete, the Proca sector should be regarded as an effective ansatz whose parameters (m_s , ξ , g_ϵ) are fixed by the framework constants and the zero-state vector, rather than derived from the substrate.

We emphasize this explicitly because the paper’s other derivations (the equation of state, the bulk viscosity, the Proca mass) are complete within the framework, whereas the Proca sector itself is not. The referee’s concern about the tension between “derived from first principles” and “assumed EFT ansatz” is well taken, and the correct resolution is to state the Proca sector’s status clearly: it is an effective framework, not a substrate-derived result.

9 The Proca mass prediction

9.1 The universal failure of UV candidates

The three UV candidates predict Proca masses:

All three fail the LIGO/Virgo bound [16] by 46–48 orders of magnitude. The failure is *structural*, not parametric: the UV scale drops out entirely.

Candidate	m_s (lattice units)	m_s (eV, M_P cutoff)
(i) Spectral gap	$\sqrt{\Delta\lambda} \approx 0.765$	9.3×10^{27}
(ii) Winding	$1/L = 0.125$	1.5×10^{27}
(iii) Substrate	$1/N = 0.0156$	1.9×10^{26}

Table 1: UV predictions for the Proca mass. All three fail the LIGO/Virgo bound by 46–48 orders of magnitude.

9.2 The EDF running

The EDF mechanism of ENTROPIA-Core [3] runs the Proca mass from UV to IR via an Ohmic hidden bath:

$$\frac{dm^2}{d\lambda} = -2\gamma_f m^2. \quad (46)$$

The solution is

$$m_s(\lambda) = m_s(0) e^{-\gamma_f \lambda}. \quad (47)$$

9.3 The exponent

The total coarse-graining extent is

$$\Delta\lambda = \frac{d+1}{d-1} N = \frac{5}{3} \cdot 64 = 106.667. \quad (48)$$

The ratio $(d+1)/(d-1) = 5/3$ is the inverse of the CKM phase amplitude $a = (d-1)/(d+1) = 3/5$ [11].

9.4 The prediction

With $\gamma_f = 1$ (the EDF overdamped limit; see §9.5) and candidate (iii) UV boundary $m_s(0) = M_P/N$,

$$m_s = \frac{M_P}{N} \exp\left(-\frac{d+1}{d-1} N\right) = \frac{M_P}{64} \exp\left(-\frac{5}{3} \cdot 64\right) = 9.03 \times 10^{-21} \text{ eV}. \quad (49)$$

This is inside the LIGO/Virgo dark-photon window and just below the cosmological coherence bound at 10^{-20} eV. The required exponent to hit 10^{-20} eV exactly is 106.565, which matches $5N/3 = 106.667$ to 0.1%.

9.5 Fixed-point status of $\gamma_f = 1$

A referee has asked whether $\gamma_f = 1$ is a fixed-point theorem of ENTROPIA-Core or an empirical input. The honest answer is: *it is neither a fully derived fixed-point theorem nor a fitted parameter, but a normalization convention that has a natural physical interpretation.*

The EDF derivation [10] establishes that the effective friction coefficient γ_f enters the equation of motion as

$$m\sigma''(\ell) + \gamma_f \sigma'(\ell) + k\sigma(\ell) = \xi(\ell), \quad (50)$$

with ξ satisfying the fluctuation-dissipation relation

$$\langle \xi(\ell) \xi(\ell') \rangle = 2\gamma_f T_b \delta(\ell - \ell'). \quad (51)$$

The overdamped limit gives a linear relaxation with rate

$$\beta = \frac{k}{\gamma_f}. \quad (52)$$

In ENTROPIA-Core, the stiffness k is identified with the curvature of the free energy at the DME equilibrium: $k = 2TS_0\gamma$. Substituting into β and solving for the coarse-graining extent required to reach the observed IR mass gives

$$\Delta\lambda = \frac{\beta}{2\gamma} N_{\text{sites}}, \quad (53)$$

where $N_{\text{sites}} = N = 64$.

The choice $\gamma_f = 1$ corresponds to measuring the coarse-graining parameter λ in units of the natural dissipative scale $\tau_f \equiv \gamma_f^{-1}$. In these units, the friction coefficient is 1 by definition, and the friction time is 1 by definition. This is a normalization convention, analogous to setting $\hbar = 1$ or $c = 1$.

Two ways to close this are under investigation:

(A) Fixed-point theorem. Show that the Ohmic bath in the EDF framework has a stable IR fixed point at $\gamma_f = 1$ in the same units in which the DME stiffness is $k = 2TS_0\gamma$. This would promote $\gamma_f = 1$ from a convention to a theorem. Preliminary analysis of the EDF flow equations suggests this is plausible but not yet proven.

(B) Alternative derivation of the exponent. Show that the exponent $5N/3$ arises from a mechanism independent of the $\gamma_f = 1$ convention. For example, a topological suppression of the form $\exp(-C \cdot N)$ where C is a fixed topological invariant would give the same prediction without invoking γ_f . Whether $C = 5/3$ can be derived from the CKM phase amplitude inverse is an open problem.

Until one of these is established, the Proca mass prediction (49) should be regarded as an EDF-motivated proposal rather than a theorem. The prediction is sharp and falsifiable, but its theoretical status is that of a conjecture consistent with the EDF framework.

10 Einstein equations and backreaction

The entanglement equilibrium condition [6, 7],

$$\delta S_{\text{ent}} = \delta \langle K \rangle, \quad (54)$$

with entropy current J_S^μ and holographic saturation $S \leq \text{Area}/(4G)$, yields

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu} + \Delta_{\mu\nu}, \quad (55)$$

with the backreaction tensor

$$\Delta_{\mu\nu} = \beta(\nabla_\mu \nabla_\nu \Theta - g_{\mu\nu} \square \Theta) + \gamma H_{\mu\nu}, \quad \Theta = \sqrt{R}. \quad (56)$$

The Fisher metric $g_{\mu\nu}$ is derived from the Bures distance on the QIN state manifold [8].

11 The H_0 boost

The H_0 boost formula is

$$H_{\text{local}} = H_{\text{CMB}} \left(1 + \frac{\gamma - 0.5}{2\pi} \right). \quad (57)$$

With $\gamma = 0.702839\dots$, the boost is 3.23%, matching the KBC void depth. This is an *observed* boundary condition, not a derived quantity. The WIN overview [1] acknowledges that the void depth is likely not derivable from DME alone.

12 Falsifiability

12.1 DESI DR2 + Euclid

The equation of state $w = -1.014054\dots$ is consistent with DESI DR2 at $\Delta\chi^2 = 0.22$ [14]. The combined DESI DR3 + Euclid program is expected to distinguish $w = -1$ from $w = -1.014$ at $> 3\sigma$ through two independent channels:

- **Distances:** $w(z)$ from BAO (DESI DR3).
- **Growth:** $\zeta H/\rho$ from $f\sigma_8(z)$ (Euclid).

12.2 Proca mass detection

The Proca mass $m_s = 9.03 \times 10^{-21}$ eV is testable by:

- **CASPER-Wind** (femtotesla magnetometry, 10^{-22} – 10^{-17} eV)
- **Eöt-Wash torsion balance** (10^{-23} – 10^{-18} eV)
- **DM Radio** (10^{-13} – 10^{-9} eV)

A detection in this window confirms the WIN mechanism. Definitive closure of the window falsifies it.

12.3 Failure modes

Test	Prediction	Falsification
DESI DR3 + Euclid	$w = -1.014 \pm 0.007$	$w = -1$ at $> 3\sigma$
Euclid growth	$\zeta H/\rho = 0.00468$	$\zeta H/\rho = 0$ at $> 3\sigma$
Proca mass	$m_s \approx 9 \times 10^{-21}$ eV	Window closed at $> 3\sigma$
H_0 boost	3.23% local enhancement	No local H_0 enhancement

Table 2: Falsifiability map for ENTROPIA-EFT.

Item	Status	Source
DME strip integral $I(\alpha)$	Derived	DME [2]
Stability coefficient γ	Derived	DME [2]
Entropy well $S/S_0 = 1 - \gamma\epsilon^2$	Derived	DME [2]
Gradient flow $d\epsilon/d\lambda = -2\gamma\epsilon$	Derived	Core [3]
$ O_0 = 2$	Derived	Torus D_4
Loop factor $1/(4\pi)$	Standard QFT	2D bubble
$w + 1 = - O_0 g_\epsilon^2/(4\pi m_\epsilon^2)$	Derived (structural)	This paper
Bulk viscosity $\zeta H/\rho = (w + 1)/3$	Derived	Trace anomaly
Proca mass $m_s = M_P/N \cdot e^{-5N/3}$	Derived (EDF, conditional)	This paper
$U(1)$ origin	Candidate (iii)	Under investigation
$g_\epsilon = (1 - \gamma)m_\epsilon$	Fixed to 0.13%	Zero-state vector
Proca Lagrangian A1	Assumed	EFT ansatz
Steady-state flux A2	Assumed	EFT ansatz
Einstein equations	Derived (Jacobson)	[6, 7]
Fisher metric	Derived (Bures)	ENTROPIA [8]
Backreaction $\Delta_{\mu\nu}$	Assumed	EFT ansatz
H_0 boost	Observed	KBC void
$\gamma_f = 1$ in EDF running	Normalization convention	§9.5

Table 3: Complete assumption audit for ENTROPIA-EFT. Items marked **Assumed** are EFT ansätze awaiting derivation from the substrate.

13 Assumption audit

14 Open problems

14.1 The coupling g_ϵ

The coupling $g_\epsilon = (1 - \gamma)m_\epsilon$ is fixed to 0.13% precision by the zero-state vector. The reported value $s_\nu = -0.105198$ gives $2\sqrt{2}|s_\nu| = 0.297540$, which is 0.13% above $1 - \gamma$. The predicted value $s_\nu = -(1 - \gamma)/(2\sqrt{2}) = -0.105062$ gives the coupling exactly.

Reconstructing the 14×19 measurement matrix at higher precision would settle whether the exact nullspace value is 0.105062 or 0.105198.

14.2 The $U(1)$ origin

Of the three candidate origins for the $U(1)$ symmetry, candidate (iii) — phase symmetry of the 14-mode multiplet — is the most natural because the $U(1)$ acts on the same multiplet that produces $|O_0| = 2$. A first-principles derivation from the substrate is still open.

14.3 The EDF friction coefficient

The EDF running requires $\gamma_f = 1$ to reproduce the Proca mass. Whether this is a fixed-point theorem, an alternative derivation of the exponent $5N/3$ independent of γ_f , or a normalization convention is an open problem (see §9.5).

14.4 The H_0 boost

The H_0 boost is observed but not derived. The WIN overview [1] suggests that a first-principles derivation may be impossible from DME alone.

15 Conclusion

We have derived ENTROPIA-EFT from DME, Core, and the 8×8 torus. The equation of state $w = -1.014054\dots$, the bulk viscosity $\zeta H/\rho = 0.004685\dots$, and the Proca mass $m_s = 9.03 \times 10^{-21}$ eV all follow from the framework constants γ , N , and d . The loop factor $1/(4\pi)$ is the standard 2D mass-regulated bubble. The multiplicity $|O_0| = 2$ is the D_4 orbit size of the two top-capacity neutrino modes. The exponent $5N/3$ is the inverse CKM phase amplitude times the substrate size.

The equation of state is consistent with DESI DR2 at $\Delta\chi^2 = 0.22$ and will be tested at $> 3\sigma$ by the combined DESI DR3 + Euclid program. The Proca mass sits at the frontier of current dark-photon detection experiments.

The only quantity not derived from first principles is the coupling $g_\epsilon = (1 - \gamma)m_\epsilon$, which is fixed to 0.13% precision by the zero-state vector. This is three orders of magnitude below DESI/Euclid sensitivity, so the empirical predictions are unaffected.

The Proca sector (Lagrangian A1 and steady-state flux A2) is currently an effective ansatz. Candidate (iii) for the $U(1)$ symmetry — phase symmetry of the 14-mode multiplet — is the most promising route to a first-principles derivation.

The EDF friction coefficient $\gamma_f = 1$ is a normalization convention rather than a theorem. Two ways to close this are under investigation: a fixed-point theorem for $\gamma_f = 1$, or an alternative derivation of the exponent $5N/3$ independent of γ_f .

ENTROPIA-EFT is the vacuum sector of the WIN Paradigm. It connects directly to the WIN baseline at $w = -1$ and to DME through γ . The chain is:

$$\text{DME } \gamma \rightarrow \text{Core } L[\epsilon] \rightarrow \text{EFT } w + 1 = -\frac{(1 - \gamma)^2}{2\pi} \rightarrow \text{WIN baseline } w = -1. \quad (58)$$

WIN supplies the -1 ; DME supplies the deviation. The two combine to give $w = -1.014054\dots$

A Numerical values

Quantity	Value
$I(2)$	0.40092988263220389628003875385983806
$I''(2)$	+0.1408946727052701891991682833423190
γ	0.70283946799007245929257819650854315
$1 - \gamma$	0.29716053200992754071
$(1 - \gamma)^2$	0.088304381784029...
w	-1.0140540788576648688...
$\zeta H/\rho$	0.004684692952555...
m_s	9.03×10^{-21} eV
G_{vac}	7/128
$ O_0 $	2
$ O_1 $	8
$ O_2 $	4

Table 4: Complete numerical values of the ENTROPIA-EFT derivation. Digits beyond the empirically relevant precision are shown for reproducibility only; see §2.4 for the precision convention.

B Relation to other WIN layers

Layer	Role	Reference
FDLM/WIN	Substrate and baseline $w = -1$	[1]
DME	Stability coefficient γ	[2]
ENTROPIA-Core	Dynamical stability	[3]
ENTROPIA-EFT	Vacuum EFT (this paper)	—
ENTROPIX/MESA	Master selection	[9]
SYK/QIN	Microscopic model	[13]

Table 5: Relation of ENTROPIA-EFT to other WIN layers.

C Reproducible verification

The derivation is fully reproducible from the inputs listed in Appendix ???. The reference Python implementation performs the following steps:

1. Compute $I(\alpha)$, $I'(\alpha)$, $I''(\alpha)$ from the closed form (3) using `mpmath` at 50-digit precision.
2. Compute $\gamma = I''(2)/(2I(2))$.
3. Compute $w + 1 = -(1 - \gamma)^2/(2\pi)$.
4. Compute $\zeta H/\rho = (1 - \gamma)^2/(6\pi)$.
5. Compute $m_s = M_P/N \cdot \exp(-5N/3)$.
6. Verify the 8×8 torus spectrum: 1 zero mode, 14 at $\lambda = 4$, 49 excited.
7. Verify the D_4 orbit sizes: $|O_0| = 2$, $|O_1| = 8$, $|O_2| = 4$.
8. Verify the mass-regulated torus bubble against the continuum $1/(4\pi m^2)$ to 0.1%.

The complete reference implementation is available in the WIN Paradigm companion repository.

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